

PADDLEOCR LCD DISPLAY RECOGNITION

Evaluation findings, implementation direction, annotation plan, and next-step benchmark

Document date	15 September 2026
Current decision	PaddleOCR only for text detection and recognition
Input scope	Perfectly cropped, upright electricity-meter LCD images
Primary objective	Extract visible text while excluding graphical symbols from OCR output

Decision summary
YOLO is excluded from the selected text pipeline. PaddleOCR will perform text detection and recognition. The existing symbol-recognition subsystem remains separate and may be merged later through a shared coordinate-space output.

Prepared from the technical discussion and hands-on tests completed on 15 September 2026.

1. Executive Summary

The evaluation began with a question about freely available OCR for seven-segment LCD displays. Generic OCR, classical OpenCV methods, a pretrained YOLO + CRNN repository, Roboflow deployment, and PaddleOCR were considered. The final direction is to use PaddleOCR only for text extraction, with custom fine-tuning if the pretrained recognizer does not generalize sufficiently.

The successful legacy PaddleOCR test produced usable bounding boxes and confidence scores. On one representative LCD image, PaddleOCR recognized the main numerical reading as 0000000.6 with approximately 0.996 confidence, L1 L2 with approximately 0.998, and kW h with approximately 0.982. It also generated false or uncertain text on small segmented indicators, including D, 1C0M, and T. This shows that the pipeline is technically viable, but confidence alone is not a sufficient correctness measure.

Recommended immediate action

- Create a repeatable PaddleOCR benchmark on 50 to 100 representative LCD crops.
- Measure exact transcription accuracy, not only OCR confidence.
- Use visual overlays to verify bounding boxes and recognized strings.
- Fine-tune PaddleOCR recognition on complete text-line crops if recurring segmented-font errors remain.
- Exclude graphical symbols from OCR annotations and OCR output filtering.

2. Agreed Scope and Constraints

Area	Agreed requirement
Input	Complete crop of active LCD glass with a small consistent margin.
Orientation	Upright and straight.
Display technology	7-segment and 14-segment LCDs.
Text content	Digits, decimal points, colons, signs, characters, fixed labels, and engineering units.
Symbols	Excluded from PaddleOCR text output; handled separately if required.
Generalization	Must work across display layouts and meter families, without fixed coordinate templates.
Out of scope	Raw-meter localization, perspective correction, blur handling, and camera-condition robustness.

The broader project description requires understanding meaningful LCD content from ideal, pre-cropped inputs and emphasizes generalization across display designs. It also states that symbols and text are distinct content types and that exact reading-level accuracy matters.

3. Approaches Tested and Key Findings

Approach	Observed outcome	Decision
EasyOCR	Read some normal labels, but failed on segmented values and confused glyphs.	Not selected.
OpenCV / templates	Previously tested and judged unsatisfactory for the required generalization.	Not selected.
OICWS YOLO + CRNN	Weights loaded, but the pretrained detector did not detect the target LCD, including on cropped inputs.	Not selected.
Roboflow	Model workflow was testable, but deployment was a paid service.	Not selected.
PaddleOCR 3.7 + PaddlePaddle 3.3.1	Model download succeeded, but Windows CPU inference failed with PIR / oneDNN attribute errors.	Current stack avoided.
PaddleOCR 2.8.1 + PaddlePaddle 2.6.2	Worked in a fresh Python 3.10 virtual environment after models were downloaded.	Selected test baseline.

Environment that worked

Python 3.10.11
PaddleOCR 2.8.1
PaddlePaddle 2.6.2
CPU inference
MKL-DNN disabled

The first model download timed out on the PaddleOCR host. Once connectivity was available, the model initialized and inference completed successfully.

4. Current PaddleOCR Results

Detected text	Confidence	Assessment
0000000.6	0.99596	Main seven-segment reading detected correctly in the tested image.
L1 L2	0.99759	Correct normal-text detection.
kW h	0.98234	Correct glyphs with an inserted space; safe unit normalization may convert to kWh.
T	0.87233	Potential status text or indicator. Requires ground-truth review.
1C0M	0.65829	Likely misrecognition of stylized segmented content.
D	0.59115	Likely false OCR detection on a small graphical indicator.

Critical finding: high confidence does not guarantee a correct LCD transcription. The benchmark must compare the predicted string with manually verified ground truth.

5. Selected PaddleOCR Architecture

The selected design uses PaddleOCR directly on the already-cropped LCD image. No YOLO component is included in the text pipeline.

Processing flow

1. Load the perfectly cropped LCD image.
2. Run PaddleOCR text detection.
3. Run PaddleOCR text recognition for each detected region.
4. Preserve quadrilateral coordinates and confidence scores.
5. Apply conservative vocabulary and whitespace normalization.
6. Group text boxes into visual reading order.
7. Exclude graphical symbols from OCR text results.
8. Export raw and normalized text in structured JSON.

Recommended output fields

Field	Purpose
raw_text	Unmodified PaddleOCR transcription.
normalized_text	Only approved deterministic normalization, such as kW h to kWh.
confidence	Recognition confidence returned by PaddleOCR.
polygon	Four-point text-region coordinates.
type	text; symbols remain outside the PaddleOCR output.
review_required	True for low confidence, invalid vocabulary, or ambiguous segmented content.

Illustrative JSON

```
{ "image": "lcd_001.jpg", "detections": [ { "type": "text", "raw_text": "kW h", "normalized_text":  
"kWh", "confidence": 0.9823, "polygon": [[1362,365],[1613,365],[1613,486],[1362,486]],  
"review_required": false } ] }
```

6. Annotation Strategy for PaddleOCR

Because PaddleOCR is now the only text model, annotations should be prepared for text detection and text recognition. Individual digit boxes are not the default annotation unit.

Detection annotations

- Draw one quadrilateral around each visually continuous text line or logical text region.
- Transcribe the exact visible sequence, including leading zeros, repeated digits, decimal points, and colons.
- Examples: 0000000.6, 01:00:00, L1 L2, EP, kWh, kVAh.
- Do not annotate bell, battery, relay, signal, tamper, or other graphical icons as OCR text.
- Use an ignore or ambiguity flag when a human annotator cannot confidently read the display.

Recognition annotations

Create tightly cropped recognition images, with one crop per complete sequence:

```
recognition/reading_0001.jpg 0000000.6 recognition/reading_0002.jpg 01:00:00  
recognition/token_0001.jpg L1 L2 recognition/unit_0001.jpg kWh recognition/token_0002.jpg EP
```

Ground-truth rules

- Annotate what is visibly displayed, not what the meter is expected to show.
- Do not fix missing segments in the label.
- Do not normalize spaces or capitalization in training ground truth.
- Keep normalization rules in a separate post-processing layer.
- Preserve repeated characters because they are a major OCR failure mode.

7. Benchmark and Evaluation Plan

Before fine-tuning, benchmark the pretrained model on a representative set. The goal is to determine whether errors are isolated or systematic.

Pilot dataset

Split	Suggested size	Purpose
Pilot	50-100 images	Validate OCR pipeline, visualization, transcription rules, and annotation format.
Training seed	200-300 images	First recognition fine-tuning experiment if pretrained exact accuracy is insufficient.
Validation	15-20%	Model selection; separate by meter family or session.
Final test	Held-out families	Measure generalization without repeated tuning.

Primary metrics

- Exact full-string accuracy for numerical readings.
- Character error rate for diagnostic analysis.
- Decimal-point recall and colon recall.
- Unit exact-match accuracy.
- False-text rate on graphical symbols.
- Accuracy by meter/display family.
- Manual-review rate after confidence and vocabulary filtering.

Required stress cases

- Leading zeros: 0000123.4
- Repeated digits: 0000000.6, 111111.1, 888888.8
- Time and date separators: 01:00:00
- Decimal placement: 0206.32 versus 02063.2
- Small segmented letters or status codes
- Different LCD colors, polarities, segment thicknesses, and spacing
- Blank, partially lit, and ambiguous positions

8. Conservative Post-processing

Post-processing should improve formatting without inventing content. Keep both raw and normalized values.

Permitted examples

```
"kw h" -> "kWh" "KV A h" -> "kVAh" leading/trailing whitespace -> removed
```

Not permitted

- Adding a decimal point or colon that PaddleOCR did not detect.

- Replacing an uncertain letter with a digit solely because a number is expected.
- Deleting repeated zeros because the output appears too long.
- Converting an icon into text.
- Using fixed coordinates as the primary recognition method across meter families.

Initial review rule

Flag a result for review when confidence is below a chosen threshold, the text violates the approved vocabulary or numeric pattern, the detected region overlaps a known symbol region, or the normalized text differs substantially from the raw OCR result.

9. Immediate Next Steps

Step	Action	Deliverable
1	Freeze the PaddleOCR legacy environment that works.	requirements.txt or pip freeze file.
2	Run OCR plus visual overlays on 50-100 representative images.	Images with boxes, text, and confidence.
3	Create verified ground truth for every test image.	CSV or JSON manifest.
4	Calculate exact reading accuracy and error categories.	Baseline evaluation report.
5	Build recognition crops for recurring failures.	PaddleOCR recognition dataset.
6	Fine-tune the recognizer if baseline accuracy is insufficient.	Custom PaddleOCR weights.
7	Re-run held-out evaluation by meter family.	Generalization report.
8	Integrate structured text output with the existing symbol subsystem if required.	Unified JSON output.

Final decision

Continue with PaddleOCR as the sole text-detection and text-recognition technology. Use the working PaddleOCR 2.8.1 / PaddlePaddle 2.6.2 environment for the immediate benchmark. Fine-tune PaddleOCR on exact LCD text-line crops only if the pretrained model fails the agreed exact-match target.

Basis documents

This report also aligns the discussion with the uploaded project documents: LCD Display Recognition System - Problem Definition, Revision 2 and LCD Display Recognition System - SDS v1.1 - Text + Symbols. Where those documents specify a YOLO text baseline or digit-wise detector annotations, this report records the new project decision to replace that text path with PaddleOCR line/region detection and recognition.